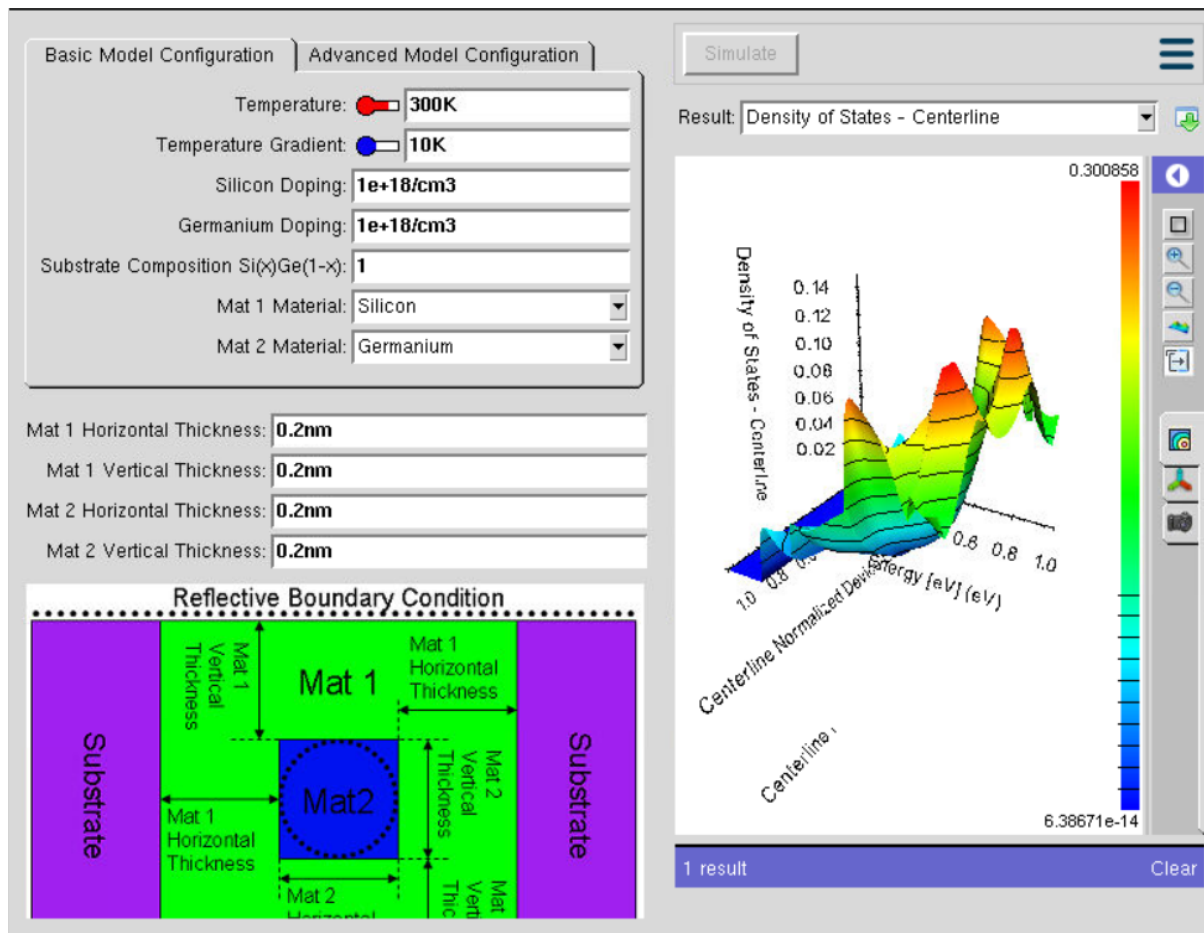
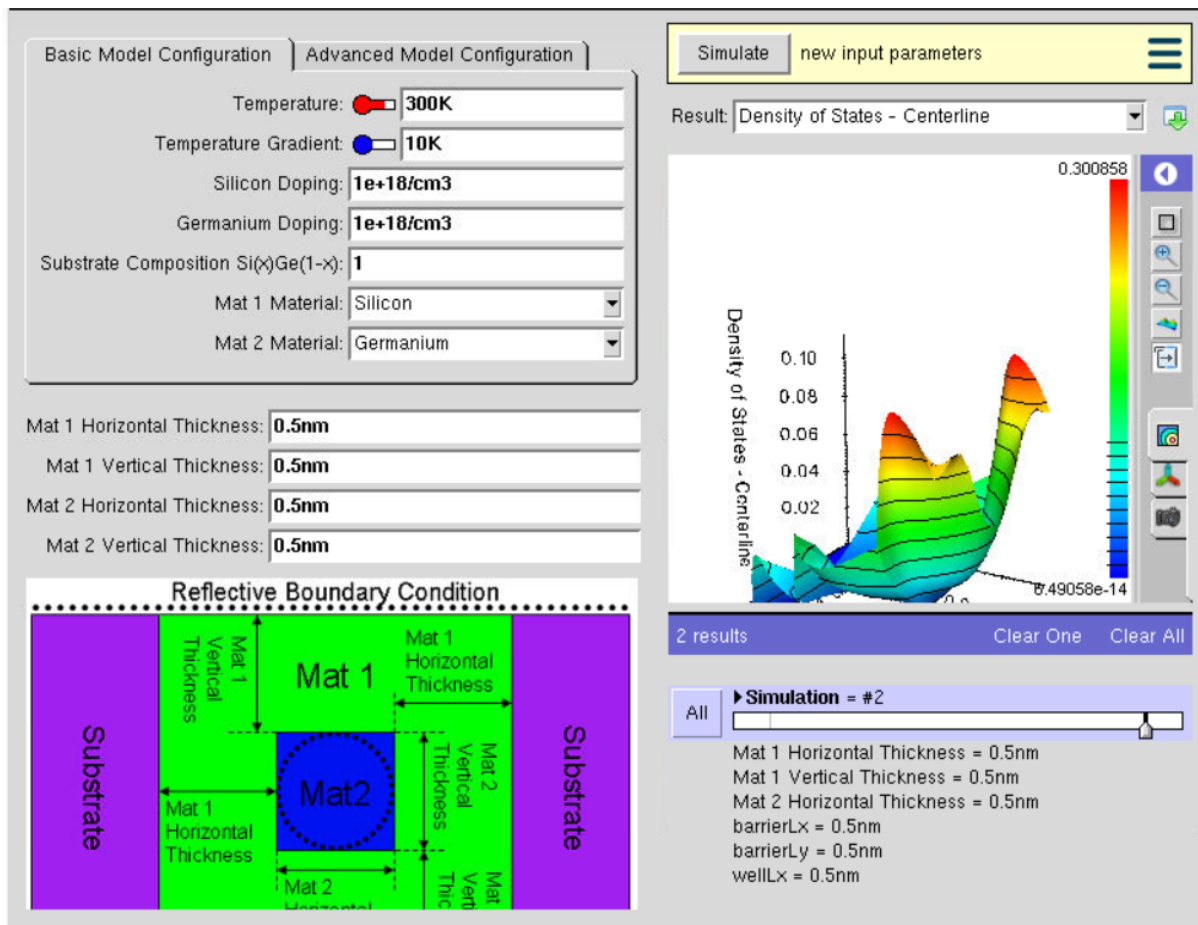


Exp-2 Simulation of Density of States of Quantum Well Structure with Various Nanometer Thickness.



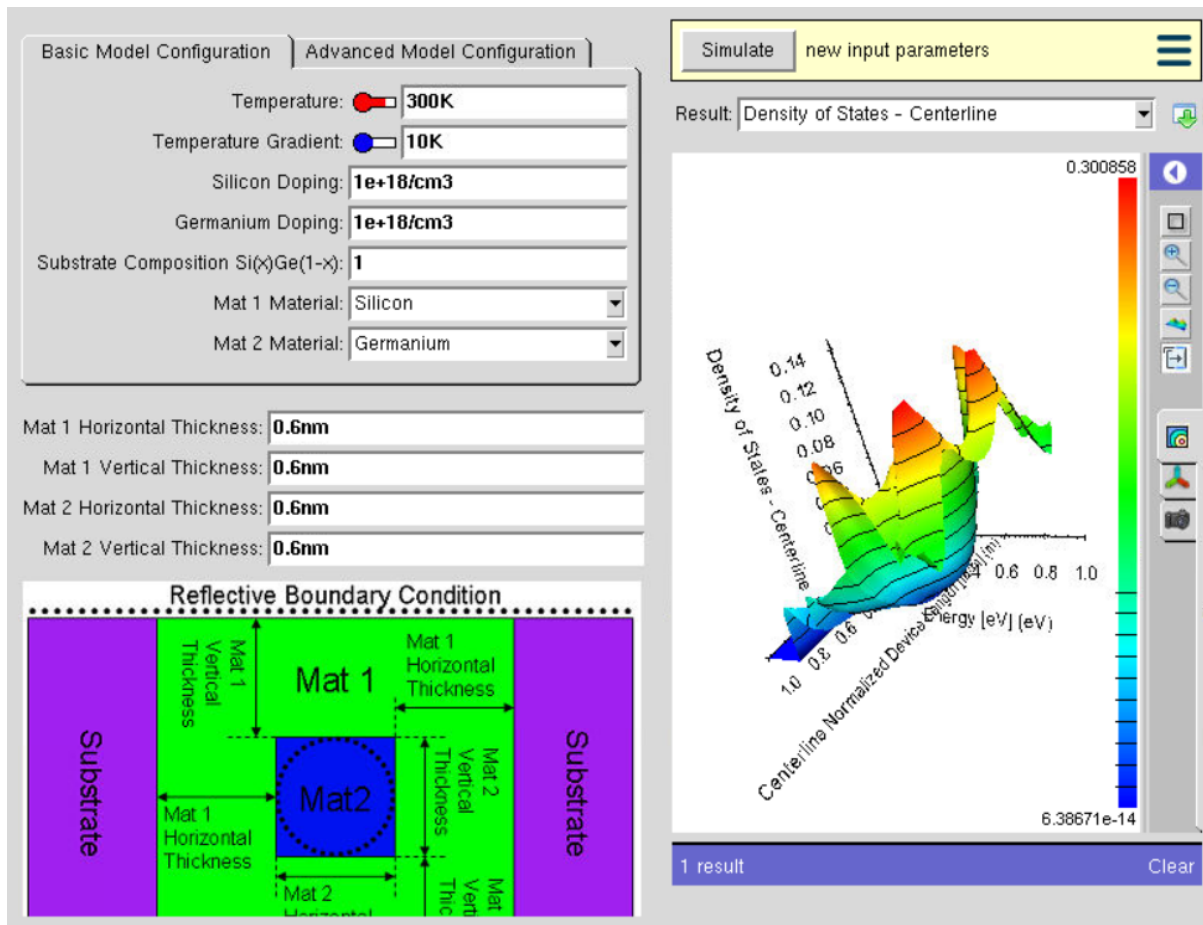
This image shows the same **Nanoelectronics simulation** of a **Silicon–Germanium (Si–Ge) heterostructure**, but with the **horizontal and vertical thicknesses of both materials increased from 0.2 nm to 0.5 nm** while keeping the temperature (**300 K**), temperature gradient (**10 K**), and doping concentrations ($1 \times 10^{18} \text{ cm}^{-3}$) unchanged. After clicking **Simulate**, the software generates the **Density of States (DOS) – Centerline** graph, which displays the distribution of available electron energy states in the modified device. Compared to the previous simulation, the larger material dimensions change the DOS profile, resulting in different peak heights and distributions due to altered quantum confinement and carrier behavior. The results list at the bottom confirms the updated thickness values used in Simulation #2. This simulation helps researchers study how changing the physical dimensions of Si–Ge nanostructures affects their electronic properties, enabling the optimization of high-speed, low-power semiconductor devices before fabrication.

Exp-2 case study-1: Simulate the density of states of quantum well of thickness 0.5 nm



This image shows the **third Nanoelectronics simulation** of a **Silicon–Germanium (Si–Ge) heterostructure**, where the **horizontal and vertical thicknesses** of both Silicon (Mat 1) and Germanium (Mat 2) have been increased to **0.6 nm**, while the **temperature (300 K)**, **temperature gradient (10 K)**, and **doping concentrations ($1 \times 10^{18} \text{ cm}^{-3}$)** remain unchanged. The device structure with reflective boundary conditions is shown on the left, and after running the simulation, the software generates the **Density of States (DOS) – Centerline** graph on the right. Compared to the previous simulations (0.2 nm and 0.5 nm), the increased thickness further modifies the distribution and intensity of the DOS peaks, indicating changes in the available electron energy states due to reduced quantum confinement and altered carrier transport. This analysis helps researchers understand the impact of device dimensions on the electrical characteristics of Si–Ge nanostructures and optimize their design for high-performance, low-power semiconductor devices before fabrication.

Exp-2 case study-3 Simulate the density of states of quantum well of thickness 0.6 nm



This simulation illustrates the **Density of States (DOS)** of a **Silicon–Germanium (Si–Ge) heterostructure** after increasing the thickness of both materials to **0.6 nm**. The 3D DOS plot shows a more pronounced distribution of energy states, with higher peaks indicating regions where electrons are more likely to occupy available energy levels. Compared to smaller device dimensions, the increased thickness reduces quantum confinement effects and allows better carrier movement within the semiconductor structure. The reflective boundary conditions ensure that charge carriers remain confined within the simulation region, providing more accurate nanoscale analysis. These results help evaluate the electrical behavior, carrier transport characteristics, and energy distribution of the device, enabling researchers to optimize the structural dimensions for improved performance, lower power consumption, and enhanced reliability in advanced nanoelectronic applications such as high-speed transistors and integrated circuits.